

Title of the thesis:

Prague Mapped by People and Satellites

Assignment of the thesis:

The aim of the thesis is to create an interactive and user-friendly visualization of an open dataset from participatory emotional mapping and to explore possibilities for integrating Copernicus Earth Observation data into the spatial analysis and visualization process.

The dataset, collected in Prague in 2021 via the EmotionalMaps.eu platform, contains over 98,000 geolocated points and more than 30,000 citizen comments. The student will preprocess and analyze the dataset to uncover spatial patterns and trends that are not immediately visible, using appropriate GIS and statistical methods. A key part of the work will be the exploration of how Copernicus data—such as Sentinel-2 imagery, urban heat islands, or land cover information—can enhance the understanding of these perceptions, particularly in relation to environmental quality and urban resilience.

Based on the findings, the student will design and implement an interactive web-based map or application that visualizes both the participatory and satellite data in a coherent, comprehensible, and visually engaging way. The application should allow filtering by theme, location, or demographic characteristics and should integrate textual comments from respondents.

An important component of the work will be the evaluation of the user interface through usability testing, ideally involving both lay users and professionals from urban planning. The outcome of the thesis will be a fully functional interactive visualization.

The student will create a website about the thesis following the rules available on the department's website and a poster about the diploma thesis in A2 format. The student will submit the entire text (text, attachments, poster, outputs, input, and output data) in digital form on a storage medium and the text of the thesis in two bound copies to the secretary of the department.

The extent of graphic works:

as required

The extent of the thesis:

maximum 40 pages of text excluding appendices

List of recommended literature:

Pánek, J., Barvíř, R., Koniček, J., & Brlík, M. (2021). The emotional map of Prague—data on what locals think about the Czech capital?. *Data in brief*, 39, 107649.

Pánek, J. (2011, September). Participatory and Public Participation GIS: A Phenomenon of Neocartography with a High Potential in Developing Countries. In 5th International Conference on Globalization, Czech Republic.

Květoňová, V., Pánek, J., Geletič, J., Šimáček, P., & Lehnert, M. (2024). Where is the heat threat in a city? Different perspectives on people-oriented and remote sensing methods: The case of Prague. *Heliyon*, 10(16).